

Armaan Jass

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EDUCATION

University of Guelph

Sept 2024 – April 2029

Bachelor of Engineering, Mechanical Engineering

Guelph, ON

- Relevant Coursework: Eng Design I–II, Materials Science, Mechanics I–II, Machine Design, Thermodynamics
- Dean's Honour List (Winter/Fall 2025, Winter 2026); Overall GPA: 3.7

WORK EXPERIENCE

Hydraulic/Electrical Technician, Angus Irrigation

May 2026 – August 2026

Hamilton, ON

- Troubleshooting & Diagnostics: Troubleshot irrigation control and hydraulic systems by tracing **electrical circuits**, **control wiring**, solenoids, valves, and **hydraulic lines** to isolate faults and determine root causes. Applied systematic problem-solving and **real-time diagnostics** to identify failures and implement targeted repairs while minimizing **downtime**, unnecessary component replacement, and repair costs.
- Electrical & Control Systems: Installed and rewired irrigation **control circuits**, including controllers, field wiring, and solenoid-operated valves. Developed practical experience with **electrical troubleshooting**, circuit tracing, **continuity testing**, **control systems**, and wiring configurations to diagnose and restore system operation.
- Hydraulic System Design & Optimization: Designed and configured irrigation systems based on **water pressure**, **flow rate**, coverage area, sprinkler specifications, zone requirements, and operating time. Applied **hydraulic calculations** to optimize water distribution and system coverage while balancing **cost**, **reliability**, **material usage**, and installation constraints to deliver safe and effective solutions.

Project Manager, Enabled Talent

Nov 2025 – Jan 2026

Remote

- Conducted **market research** in Ghana to identify employment opportunities, accessibility needs, and potential stakeholders, informing a theoretical expansion and **project implementation strategy** for an accessible talent enablement platform.
- Converted workforce needs, infrastructure constraints, and accessibility requirements into **functional workflows** and high-level **operational requirements**.
- Evaluated **scalability**, usability, and **deployment feasibility** to support real-world implementation in an emerging-market context.

ENGINEERING PROJECTS

Autonomous Multi-tool drone, Personal Project

May 2026 – August 2026

Cambridge, ON

- Calibrated, tested, and validated core **flight control**, motor response curves, and **power distribution** parameters utilizing **QGroundControl** and **Mission Planner** to ensure flight stability and safe **autonomous operation**.
- Wired and integrated a high-performance **power distribution system** pairing a 6S LiPo battery, **4-in-1 ESC**, and power module with a **Pixhawk 6C flight controller**, high-gain GPS, ELRS receiver, **Raspberry Pi 4**, and companion vision sensor.
- Fully designed and iterated functional drone housing structures in **Fusion 360** using **PA6 nylon-carbon fiber composite**; optimized balance and **structural rigidity** through custom diamond-saw machining, drill-press post-processing, and soldered **thermal heat-set inserts** for structural thread strength.
- Developed multiple motor mount and landing gear iterations, analyzing **structural failure points** to pivot from initial PLA designs to high-impact PA6 to **reduce weight by 8%** as well as increase **tensile and compressive strength**.

Brakes, Aerodynamics and Driveline Member, Formula SAE

Fall 2024 – Present

University of Guelph, ON

- **Reverse engineered** a mechanical throttle body using precision measuring tools and **SolidWorks** modeling; produced and validated a functional **3D-printed assembly** for vehicle testing and driving.
- Engineered brake fluid reservoir mounting hardware using **DFM/DFA** principles, accounting for **vibration loading**, fastener access, assembly sequence, serviceability, and brake line routing constraints.
- Supported vehicle fabrication through **welding operations** and **composite manufacturing**, including fiberglass and **carbon fiber hand layup** for structural and safety-related components.
- Prepared manufacturing-ready documentation including fully dimensioned **2D drawings** with **GD&T tolerance callouts** for internal assembly and external fabrication.

Design II – Mechanical Engineering Project

Fall 2025

University of Guelph, ON

- **Reverse engineered** a consumer toy using calipers and **parametric SolidWorks modeling**, recreating a full **3D assembly** and producing detailed **2D engineering drawings** and motion animations to demonstrate function and assembly.

LEADERSHIP & INVOLVEMENT

Team Captain, Provincial Soccer

2023 – Present

Oriental Sports Club - Cambridge, ON

- Led **team strategy**, **communication**, and **accountability** under high-pressure competitive environments, driving performance to secure a regional championship.